teams sometimes triggered unexpected denials because entity data was formatted differently across systems.

This experience led ACME to recognize the importance of a consistent way to represent entities. The company started designing a normalization layer—an enterprise PIP that would gather data from all internal and external sources, apply a unified schema, and deliver well-formed entities to each PDP. Later in this chapter, we'll return to that decision in the section on authorization context normalization, where we'll see how it simplified PDP logic, improved policy consistency, and made ACME's overall authorization system more reliable.

12.3 Signals and their delivery

Once entities are defined, the next step is gathering the data that populates them. The signals that populate entity attributes—who someone is, what they're trying to access, what device they're using, and what risk level they represent—come from many directions. These *trust signals* are essential for establishing the authorization context.

Different use cases draw from different combinations of signals, delivery methods, and freshness levels. Taken together, these signals determine how the system assembles authorization context at runtime.

12.3.1 What the signals are

At a minimum, every authorization system needs identifiers for principals and resources, but most real systems use richer signals. Common examples include:

- Identity and role details from directories or HR systems.
- Device posture from endpoint clients or mobile device managers.
- Account status and entitlements from customer or subscription databases.
- Environmental and risk data, such as location, time of day, or fraud scores.